Finding the Unknown in an Equation

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Equation

An **equation** is a mathematical statement that shows two expressions are equal using the equal sign ($=$).

$$x + 3 = 10$$

Left side $=$ Right side

Unknown

An **unknown** is a number whose value is not yet known.
It is usually represented by a letter such as x, n, or a.

$$x + 3 = 10$$

Here, x is the unknown.

Finding the Unknown

To find an unknown, determine the number that makes the equation true.

$$x + 3 = 10$$

$x = 7$ makes the equation true because

$$7 + 3 = 10$$

Inverse Operations

Use the **inverse operation** to undo the operation applied to the unknown.

Operation	**Inverse**
Addition	Subtraction
Subtraction	Addition
Multiplication	Division
Division	Multiplication
Squaring	Principal square root

Keep It Balanced

An equation is like a balanced scale.

Whatever you do to one side, do to the other side.

Addition and Subtraction

Isolate the unknown by performing the appropriate inverse operation on both sides.

If the equation has	**Do this to both sides**
$+$	Subtract
$-$	Add

Multiplication and Division

Use the appropriate inverse operation to isolate the unknown.

If the unknown is	**Do this**
Multiplied	Divide
Divided	Multiply

Equations with Squares

If $x^2 = a$ and x is non-negative, then

$$x = \sqrt{a}$$

provided $a \geq 0$.

$$x^2 = 25 \quad \Rightarrow \quad x = \sqrt{25} = 5$$

Examples

Example 1

Find x.

$$x + 7 = 12$$

Subtract 7 from both sides.

$$x = 12 - 7 = 5$$

Example 2

Find n.

$$n - 4 = 9$$

Add 4 to both sides.

$$n = 9 + 4 = 13$$

Example 3

Find a.

$$3a = 21$$

Divide both sides by 3.

$$a = \frac{21}{3} = 7$$

Example 4

Find m.

$$\frac{m}{6} = 4$$

Multiply both sides by 6.

$$m = 4 \times 6 = 24$$

Example 5: Two Steps

Find x.

$$2x + 5 = 13$$

Step 1: Subtract 5 from both sides.

$$2x = 8$$

Step 2: Divide both sides by 2.

$$x = 4$$

Example 6

Find x. Assume x is non-negative.

$$x^2 = 25$$

Take the principal square root.

$$x = \sqrt{25} = 5$$

Example 7

Find n. Assume n is non-negative.

$$(n+1)^2 = 16$$

Take the principal square root.

$$n+1 = 4$$

Subtract 1 from both sides.

$$n = 3$$

Practice Exercises

Practice 1

Find the value of the unknown. Assume that the unknown is non-negative.

$$x + 8 = 15$$

Practice 2

Find the value of the unknown. Assume that the unknown is non-negative.

$$n - 6 = 11$$

Practice 3

Find the value of the unknown. Assume that the unknown is non-negative.

$$4a = 28$$

Practice 4

Find the value of the unknown. Assume that the unknown is non-negative.

$$\frac{m}{5} = 7$$

Practice 5

Find the value of the unknown. Assume that the unknown is non-negative.

$$2x + 3 = 17$$

Practice 6

Find the value of the unknown. Assume that the unknown is non-negative.

$$3n - 4 = 20$$

Practice 7

Find the value of the unknown. Assume that the unknown is non-negative.

$$\frac{y}{3} + 5 = 9$$

Practice 8

Find the value of the unknown. Assume that the unknown is non-negative.

$$5a + 10 = 35$$

Practice 9

Find the value of the unknown. Assume that the unknown is non-negative.

$$x^2 = 49$$

Practice 10

Find the value of the unknown. Assume that the unknown is non-negative.

$$(n+2)^2 = 36$$

Activities

Activity 1

Find the value of the unknown. Assume that the unknown is non-negative.

$$x + 9 = 21$$

Activity 2

Find the value of the unknown. Assume that the unknown is non-negative.

$$n - 8 = 13$$

Activity 3

Find the value of the unknown. Assume that the unknown is non-negative.

$$6a = 42$$

Activity 4

Find the value of the unknown. Assume that the unknown is non-negative.

$$\frac{m}{4} = 9$$

Activity 5

Find the value of the unknown. Assume that the unknown is non-negative.

$$3x + 2 = 20$$

Activity 6

Find the value of the unknown. Assume that the unknown is non-negative.

$$4n - 5 = 23$$

Activity 7

Find the value of the unknown. Assume that the unknown is non-negative.

$$\frac{y}{2} + 6 = 14$$

Activity 8

Find the value of the unknown. Assume that the unknown is non-negative.

$$7a + 7 = 42$$

Activity 9

Find the value of the unknown. Assume that the unknown is non-negative.

$$x^2 = 64$$

Activity 10

Find the value of the unknown. Assume that the unknown is non-negative.

$$(n+3)^2 = 81$$

Thank you!